Generating Chest X-rays with a VAE

Your Name
ECE 285: Homework 1, UC San Diego

1 Method

We implement a convolutional VAE [1] in PyTorch. The encoder predicts a diagonal Gaussian $q_\phi(z \mid x)$ through mean μ and log variance ℓ. We sample $z = \mu + \exp(\ell/2) \odot \epsilon$, where $\epsilon \sim \mathcal{N}(0, I)$, and minimize

$$\mathcal{L} = \frac{1}{B} \sum_{i=1}^{B} \left[\sum_{p} (x_{ip} - \hat{x}_{ip})^2 + \frac{\beta}{2} \sum_{j} (\mu_{ij}^2 + e^{\ell_{ij}} - 1 - \ell_{ij}) \right]. \tag{1}$$

Terms are summed per image and batch-averaged. Squared error is a Gaussian likelihood loss (variance $1/2$); $\beta = 1$ gives the negative ELBO up to a constant.

Encoder blocks use kernel 4, stride 2, padding 1, LeakyReLU(0.2), and channels $(16, 32, 64, 128)$ by default. Linear posterior heads and a mirrored transposed-CNN decoder with ReLU/final sigmoid complete the model. Generation decodes prior draws into image means.

2 Experimental setup

Kaggle X-rays [2] are split into 5,216 training, 16 validation, and 624 test images, pooling classes without labels. Inputs are grayscale, square-resized, scaled to $[0, 1]$, without augmentation. CPU training uses Adam, learning rate 10^{-3}, batch size 32, and 30 epochs (64^2 baseline) or 40 (others). Checkpoints minimize validation loss. Training seeds are listed in Table 1.

Evaluation. Following Ignite [3], 624 prior samples (seed 123) and 624 test images are evaluated in batches of 16. Real originals and native generated outputs are resized to 299^2. Inputs use PIL bilinear resizing, uint8/RGB conversion, and $[0, 1]$ scaling. Frozen Inception v3 retains `transform_input=True`, without external normalization. FID uses *1,000 softmax probabilities*, not 2,048 pooled features; benchmark scales differ. IS uses the full generated set. All reported runs share the fixed real reference.

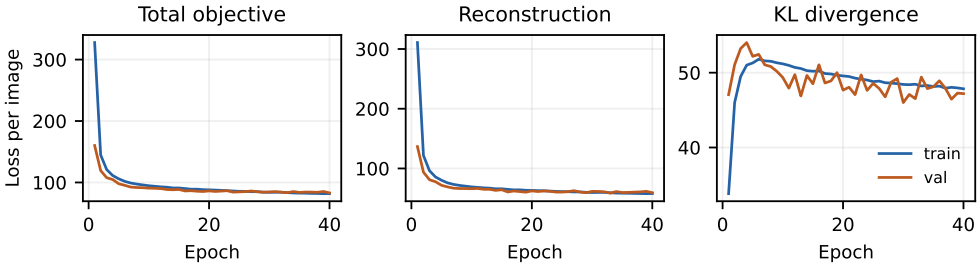


Figure 1: Training and validation losses: three blocks, 128^2, latent 32, $\beta = 0.5$, seed 42. Losses decrease, with noisy validation estimates.

3 Results and discussion

Table 1 reports selected checkpoints. MSE divides validation reconstruction sums by pixel count; cross-resolution targets still differ.

Table 1: H: image side; D: blocks; d_z: latent dimension; s: training seed; e^*: selected epoch. MSE is validation pixel MSE. Lower FID and higher IS are conventionally preferred.

H	D	d_z	β	s	e^*	MSE	FID	IS
64	4	32	1	42	23	.006006	.2351	1.4971
128	4	32	1	42	38	.004090	.2462	1.8396
128	4	64	1	42	38	.004163	.2459	1.8723
128	4	32	.5	42	32	.003562	.2440	1.9334
128	4	32	2	42	38	.004997	.2518	1.7648
128	4	32	.5	43	40	.003586	.2334	1.9352
128	4	32	1	43	33	.004195	.2766	1.9183
128	3	32	.5	42	40	.003628	**.2150**	1.9546
128	5	32	.5	42	28	.003598	.2785	**1.9675**

Resolution, latent size, and regularization. Increasing resolution does not consistently improve FID, and doubling latent dimension gives negligible benefit. Lowering β from 1 to 0.5 reduces pixel MSE by 13–15% and improves FID/IS in both seeds. At seed 42, KL increases from 32.54 to 46.20, reflecting weaker regularization. $\beta = 2$ worsens reconstruction and both generation scores. Total losses across beta values are different objectives.

Architecture. Three blocks lower FID by approximately 12% versus four blocks, despite slightly worse reconstruction. Five blocks have the highest IS but worse FID. Depths 3/4/5 yield $16^2/8^2/4^2$ final feature maps and 1.672/1.140/1.791M parameters at 128^2; this is not an isolated depth effect.

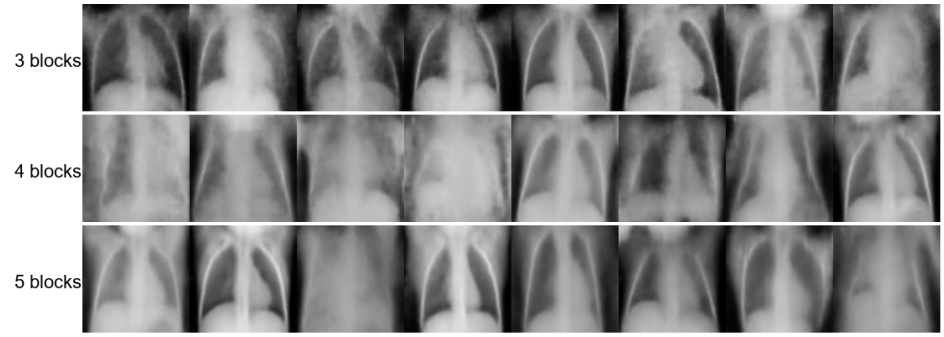


Figure 2: First eight evaluated prior draws, generation seed 123. Rows: 3/4/5 blocks; all use 128^2, latent 32, $\beta = 0.5$, training seed 42. Selected epochs: 40/32/28. Broad chest shapes vary, but detail remains blurred or patterned.

Limitations and conclusion. Validation has only 16 images; test metrics were inspected during exploration, so these are descriptive results, not an untouched holdout assessment. Two seeds support the beta comparison; depth uses one. Resolution also changes budget, capacity, and reconstruction/KL balance. Small sample counts and ImageNet features limit metric interpretation. $\beta = 0.5$ is promising; the three-block FID advantage needs seed replication and parameter-matched testing. Clinical validity is not established.

References

[1] D. P. Kingma and M. Welling. Auto-Encoding Variational Bayes. ICLR, 2014. arXiv:1312.6114.

[2] P. Mooney. Chest X-Ray Images (Pneumonia). Kaggle dataset.

[3] PyTorch-Ignite. GAN evaluation using FID and IS. Evaluation tutorial.